

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level / CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Define Big Data? What are challenges and Applications of Big Data?	1	
B) Which are the enabling technologies of Big data Explain with suitable Example.	1	
C) Explain Big Data stack with suitable example.	1	
Q.2 Solve Any Two of the following.		12
A) Write Note on: 1. Cassandra 2. HBase	2	
B) Analyze the Role of Zookeeper in brief? How information is stored in Zookeeper explain workflow of zookeeper.	2	
C) What is consistency trade-of. Write a difference between ACID and BASE Model. Which Model will be suitable for Stock Market.	2	
Q.3 Solve Any Two of the following.		12
A) Enlist and explain the GRAPH Algorithms. How is GraphX different when compared to Giraph?	4	
B) Explain K-Means clustering Algorithm. How the optimal value of K is decided.	4	
C) What is mahout ?Explain features and Applications of Mahout.	2	
Q.4 Solve Any Two of the following.		12
A) What is Data Streaming? Why is Real-time Big Data Pipeline So Important Nowadays?	3	
B) What are the Different Features of a Real-time Big Data Pipeline System?	3	
C) What is Messaging System? Explain Role and working of KAFKA.	3	
Q.5 Solve Any Two of the following.		12
A) Where we can use MongoDB? Write the Features of MongoDB.	2	
B) What is difference between MongoDB and RDBMS.	1	
C) What are NoSQL databases? What are the different types of NoSQL databases?	2	